III B.TECH – II SEMESTER

Software Project Management

Professional Elective-III | JNTUA R23 · B.Tech CSE

	SOFTWARE PROJECT MANAGEMENT (Professional Elective-III)	L	T	P	C
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Course Objective:

This course is designed to enable the students to understand the fundamental principles of Software Project management & will also have a good knowledge of the responsibilities of a project manager and how to handle them.

Course Out comes:

After completion of the course, students will be able to

- Describe the fundamentals of Project Management
- Recognize and use Project Scheduling Techniques
- Familiarize with Project Control Mechanisms
- Under stand Team Management
- Recognize the importance of Project Documentation and Evaluation

UNIT-I Lecture 9Hrs

Conventional Software Management: The water fall model, conventional software Management performance Evolution of Software Economics: software Economics. Pragmatic Software Cost Estimation Improving Software Economics: Reducing Software Product Size, Improving Software Processes, Improving Team Effectiveness, Improving Automation, Achieving Required Quality ,Peer Inspections.

UNIT-II Lecture 9Hrs

The old way and the new: The principles of convention al software Engineering, principles of modern software management, transitioning to aniter ative process.

Lifecycle phases: Engineering and production stages, inception, Elaboration, construction, transition phases.

Artifacts of the process: The artifact sets, Management artifacts, Engineering artifacts, programmatic artifacts

UNIT-III Lecture 9Hrs

Work Flows of the process: Software process work flows, Inter Trans work flows.Check points of the Process: Major Mile Stones, Minor Milestones, Periodic status assessments. Iterative Process Planning: work break down structures, planning guidelines, cost and schedule estimating, Iteration planning process, Pragmatic planning

UNIT-IV Lecture 9Hrs

Process Automation: Automation Building Blocks, The Project Environment.

Project Control and Process instrumentation: The sevencore Metrics, Management indicators, quality indicators Tailoring the Process: Process discriminants. Managing people and organizing teams.

UNIT-V Lecture 9Hrs

Project Organizations and Responsibilities: Line - of-Business Organizations, Project Organizations, evolution of Organizations.

Future Software Project Management: modern Project Profiles, Next generation Software economics, modern process stransitions.

Case Study: The Command Center Processing and Display System-Replacement(CCPDS-R)

Text books:

- Software Project Management, Walker Royce,Pearson Education,2012
- BobHughes,MikeCotterellandRajibMall“SoftwareProjectManagement”,6thEdition,

Mc Graw Hill Edition, 2017

Reference Books:

- PankajJalote,“SoftwareProjectManagementinpractice”,5thEdition,PearsonEducation,

17.

- Murali K.Chemuturi,Thomas M.Cagley Jr.”Mastering Software Project Management: Best Practices, Tools and Techniques”, J.Ross Publishing, 2010
- SanjayMohapatra,“SoftwareProjectManagement”,CengageLearning,2011

Online Learning Resources:

http://nptel.ac.in/courses/106101061/29